

Suspicious Digital Asset Investigation

a) Detecting hidden data in the image

Visual Inspection - Look for unusual pixels, color changes, or image distortions

Metadata analysis - check EXIF/file metadata for unusual comments, file size, software, or time stamps

Statistical analysis - Analyse pixel/color distributions to identify abnormal patterns caused by hidden data

LSB analysis - Examine the least significant bits of pixels because hidden information is commonly stored there

File comparison - Compare the suspicious image with the original logo to identify unexpected changes

b) Two techniques to strengthen hidden communication

1) Encrypt the hidden data before embedding

Even if the hidden information is extracted encryption prevents unauthorized users from understanding it

2) Use adaptive/randomized embedding

Hide data in less noticeable image regions and distribute it pseudo-randomly rather than sequentially, making statistical detection more difficult.

Embedding suspicious transaction
Red flags in policy

Step 1:

Step 2:

Detecting of suspicious transaction by front
line staff

Escalating to MLRO

Step 3:

Investigating internally

Step 4:

Step 5:

Filing to FIU (goAML)

Post-filing actions

Step 6:

Step 7:

Additional Reporting and Instructions

Continuous monitoring and training

Step 8:

cipher text:

Communication

we received some a cipher text

a) Recovering the original message

This is a 3-level transposition / rail-fence style arrangement
Reversing the 3-rail arrangement gives:

WE ARE DISCOVERED. FLEE AT ONCE

So the probable original message is:

WE ARE DISCOVERED. FLEE AT ONCE

b) Why is the method vulnerable?

A transposition-based encryption method only changes the order of characters and does not provide strong modern encryption. The original characters remain unchanged so patterns and frequency information can still be analyzed by an attacker. Since the number of possible arrangements is limited, modern computers can test different arrangements quickly. It also lacks strong key management and authentication mechanisms.

Therefore, this method does not provide sufficient confidentiality. For modern corporate communication

2) Practical enhancement

A practical improvement

A practical improvement would be to replace the legacy transposition method with a modern encryption algorithm such as AES-256-GCM.

AES provides authentication and integrity protection - the message should be encrypted using a securely managed secret key before transmission

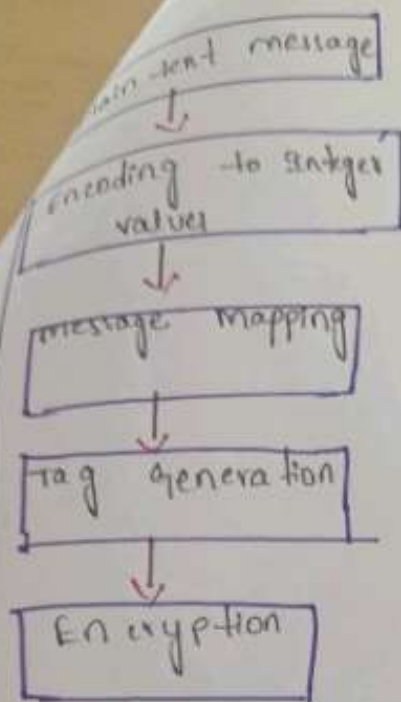
This provides:

- Strong encryption
- Key-based security
- Confidentiality
- Integrity
- Authentication

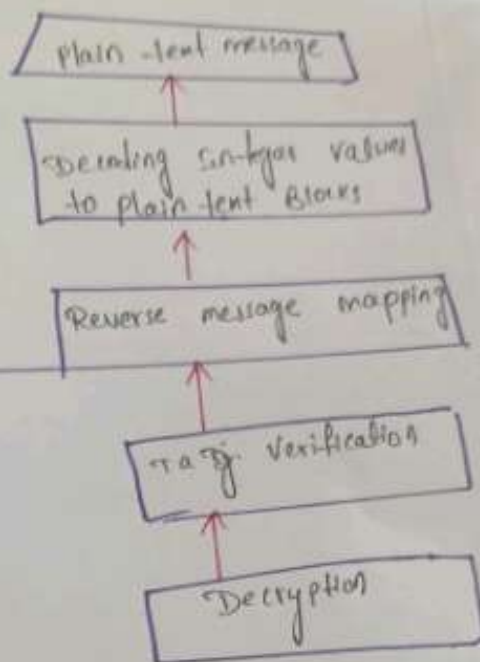
In short:

Old: transposition → rearranges letters only

Improved: AES-GCM → encrypts the actual data using a secret key



Sender side



Receiver side

Conclusion :-

Steganalysis can help the investigators identify hidden information in digital images, while encryption can protect the hidden information from unauthorized access. Similarly, replacing out data transposition encryption with modern algorithm such as AES-256-GCM provides stronger confidentiality and security for corporate communication.